

CS-350 - Emerging Sys Arch & Tech

7-1 Final Project: Thermostat Lab

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SysTec Smart Thermostat Prototype Report

The prototype developed for SysTec's smart thermostat addresses the foundational requirements of a low-level embedded thermostat system. It successfully integrates hardware peripherals to manage heating and cooling states, process user inputs, display system status, and simulate data transmission to a server. The system uses an AHT2 temperature sensor to read ambient room temperature via the I2C protocol. A red LED indicates heating, and a blue LED indicates cooling. These LEDs fade in and out when the system is actively heating or cooling and stay solid when the setpoint temperature is reached. A single button cycles between the Off, Heat, and Cool states, while two additional buttons allow the user to increase or decrease the target temperature.

Information such as the date, time, current temperature, and setpoint is shown on a 16x2 LCD display. The display alternates with the second line between showing the current temperature and the system's operating mode with its target setpoint. The device also simulates connectivity to a server by sending data over UART. This data includes the current state, room temperature in Fahrenheit, and the setpoint. A state machine was implemented in Python to manage the system's behavior, ensuring robust transitions between states and responsiveness to user input.

The system architecture was designed to support all required peripherals, including GPIO for LEDs and buttons, I2C for the temperature sensor and LCD, and UART for serial communication. The prototype was developed on a Raspberry Pi 4B, which offers abundant GPIO pins, built-in I2C and UART interfaces, and sufficient RAM and Flash memory for

handling all code and sensor inputs. The Raspberry Pi also supports Wi-Fi, making it a strong candidate for both development and production phases.

For cloud connectivity, all three architectures Raspberry Pi, Microchip (e.g., SAMD21 with Wi-Fi), and Freescale can wirelessly communicate. However, the Raspberry Pi comes with integrated Wi-Fi and a mature software ecosystem, making it easier to implement cloud services using Python-based frameworks or MQTT protocols. While lightweight and power-efficient, microchip solutions require more engineering effort to set up Wi-Fi communication and cloud integration. Freescale architectures are powerful and suitable for industrial applications but are overkill for this use case and may introduce unnecessary complexity.

All three platforms support the required peripherals and have adequate Flash and RAM. The Raspberry Pi 4B provides up to 8 GB of RAM and sufficient storage, which is far beyond what this prototype requires, but ensures future scalability. Microchip and Freescale chips offer smaller memory footprints but are highly customized for embedded applications with real-time constraints. For a cloud-connected product, the Raspberry Pi is recommended due to its developer-friendly environment, strong community support, and built-in wireless capabilities.

In conclusion, the prototype fulfills SysTec's requirements for reading temperature, controlling HVAC state indicators, responding to user inputs, and simulating server communication. Among the architectures considered, the Raspberry Pi offers the best combination of capability, ease of use, and scalability for the next phase of development: integrating full cloud connectivity into the smart thermostat.